

RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019-2020 (Odd Semester)

INNOVATIVE PRACTICE

Active Learning: Roleplay

Degree, Semester & Branch: VII Semester B.E. Computer Science and Engineering

Course Code & Title: CS6701 Cryptography and Network Security

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TOPIC: CAESAR CIPHER

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Objectives

- To expose the students to different ways of thinking about a problem, expand their ability to resolve problems.
- To understand the classical encryption techniques.
- To apply classical encryption techniques to improve the security in data (message) transactions.

Outcome (O) (At the end of the learning technique, the students will be able to)

- Interpret the classical encryption techniques to develop security applications.

Justification for choosing the topic

In the initiation of greater requirement in digital data transmission in recent time, the problem of huge damages or attacks from illegal data access has become a serious problem. Accordingly, the data security has become a serious and crucial issue in digital data transmission applications. In order to protect valuable information from unauthorized users or against illegal reproduction and modifications, various types of cryptographic schemes are needed. This course Cryptography and Network Security provide the solutions to solve this kind of issues. The objective of active learning is to expose thinking, creativity and problem-solving skills.

Active learning methodology is also a form of activity based learning. It makes all learners to participate in learning. In this method, the students involve in reading, writing, speaking, sharing, expressing the skills and questioning individually and in groups. Active learning involves students in doing things and thinking about what they are doing. In text book this chapter/topic explained in theoretical manner, so I thought to explain this topic in teach and then play manner. Since all the students wants to learning with some fun.

Meaning of Role Play

Role-play is a technique that allows students to explore realistic situations by interacting with other people in a managed way in order to develop experience and trial different strategies in a supported environment.

Definition of Role Play

Role play is an educational technique in which people spontaneously act-out problems of human relations and analyze the enactment with the help of other role players and observers.

When Should Role Playing Be Used?

- Role playing is effective when the topic involves person to person communication or interactions.
- Role play can allow everyone to participate.
- The overall mood a training session can be improved by the excitement that role playing provide.

Steps of Role Play Model of Teaching

- Selection of a problem
- Constructing of role play situation
- Casting the players
- Briefing
- Role play action
- Discussion and analysis

Caesar Cipher

The Caesar Cipher technique is one of the earliest and simplest methods of encryption technique. It's simply a type of substitution cipher, i.e., each letter of a given text is replaced by a letter some fixed number of positions down the alphabet. For example with a shift of 1, A would be replaced by B, B would become C, and so on.

$$C = (P+K) \bmod 26$$

$$P = (C-K) \bmod 26$$



In our class, students actively participated in Role play which clearly explains the Caesar Cipher. In this role play we took an example for Caesar Cipher plain text as “INDIA” and key as 3. The first student act as a character “I”, second student act as a character “N”, third student act as a character “D”, fourth student act as a character “I” and fifth student act as a character “A”. In

this example we are using Key as 3 so when we are applying the key, the first character (Student) move three position right (which means add plus three with the position), the Second character (Student) move three position right and so on.



REFLECTION CRITIQUE

Challenges

- Making student participation was challenging. Initially, the students felt shy to participate.
- The participated students did not understand the assigned roles and procedures.
- Few slow learners are not involved and understand the concepts.
- Some of the students were discussing while playing.

Steps to be taken to avoid the problems.

- Motivated the students to their involvement during the activity.
- After forming and assigning roles to the students, I explained their roles and procedures.
- I motivated the not participated members in the activity by means of benefits
- Peers also explained the concepts in the same class.

POST IMPLEMENTATION

- Based on the activity and student's feedback, they felt that it was easy to understand the concept.
- Most of the students were actively participated in Roleplay rather than one-way communication.
- After completing the Roleplay, some of the students formed their own group and conducting the same for various concepts.

RELEVANCE OF PROGRAM OUTCOMES

Programme Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Outcome(O)	3	1	1					2	2	2		2

Faculty In-charge

HOD-CSE